

Technical PoC

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QR to Game Action

1. Technical PoC Aim

The aim of this PoC is to test whether the proposed Svelte-based digital companion can interact with, represent, and respond to an event that occurs during the physical board game.

Rather than developing the complete digital companion at this stage, I focused on one small gameplay interaction: a Rat triggering a trap and losing health.

The main question being tested is:

"Can a physical component of Oh, RATS! successfully connect to the digital companion and trigger the correct digital game action?"

2. What I Implemented

I created a basic prototype using Svelte and JavaScript. From the suggested PoC options, I chose QR to Game Action:

Scan a physical trap/card → application identifies it → trap effect is triggered → Rat health is updated.

The prototype represents a Rat with 10 HP and a trap that deals 5 damage.

Initially, I tested this interaction using an Activate Trap button. When the button is pressed:

- The trap is triggered.
- The Rat loses 5 HP.
- The health changes from 10 HP to 5 HP.
- A message confirms that the trap has been triggered.
- The trap cannot be activated again.

This allowed me to first test that the basic game state and trap interaction worked correctly.

3. Physical to Digital Connection

In the physical version of Oh, RATS! the Cat player will be able to place a limited number of traps on the board. If a Rat moves onto a tile containing a trap, the trap is triggered and applies an effect to the Rat.

To test the physical to digital connection, I created a QR code containing a link to the Svelte application. When the QR code is scanned using a mobile phone, the application reads the information contained in the URL and identifies the trap. The interaction then becomes:

Physical Trap / Trap Card → Scan QR Code → Trap Identified → -5 HP → Rat Health Updates

This successfully connects a physical game component to a digital game action.

4. Development Approach

I developed the interaction incrementally rather than attempting to build the entire digital companion at once. The first stage was testing whether Svelte and JavaScript could represent and update the game's state. Once this worked, I introduced the QR code connection and tested whether the physical component could trigger the same interaction. This allowed each part of the technical approach to be tested separately before combining them.

5. QR Code Test

The QR code was tested using a mobile phone connected to the same local Wi-Fi network as the development computer.

When the QR code was scanned:

1. The Svelte application opened on the phone.
2. The application identified the trap from the URL parameter.
3. The application displayed "Trap identified!"
4. The trap automatically triggered.

5. The Rat's health changed from 10 HP to 5 HP.
6. A message confirmed that the trap had been triggered.

6. Testing Environment

The prototype was developed and tested locally using a Svelte development server. The digital companion was also accessed from a mobile phone over the same local Wi-Fi network to test the intended mobile interaction. For this PoC, the QR connection uses a local development address rather than a publicly hosted application. This was sufficient for testing whether the proposed physical to digital interaction works.

7. What Has This PoC Proven?

The PoC has demonstrated that:

- Svelte can be used to build the proposed digital companion.
- JavaScript can represent and update game state.
- A physical component can be represented using a QR code.
- The QR code can identify the relevant game component.
- Identifying the trap can trigger a game action.
- The trap can apply an effect to the Rat's health.
- The Rat's health can change from 10 HP to 5 HP.
- The digital companion can provide feedback about the game event.
- The prototype can be accessed from a mobile device.

8. Current Limitations

- The QR code currently relies on the local network.
- Only one trap interaction has been implemented.
- A final physical/3D printed trap has not yet been produced.
- The final trap placement system is still being designed.
- The application is currently a prototype rather than the final digital companion.
- The QR code currently identifies the trap through a URL parameter rather than a more advanced physical tracking system.

Local Network Limitation

The current QR code connects to the Svelte application using the computer's local network IP address. This means that, for the QR code to work, the phone scanning it and the computer running the Svelte development server must be connected to the same Wi-Fi network. For example, during testing, the computer and phone were connected to the same home Wi-Fi network. The QR code then directed the phone to the Svelte application running on the computer. If the phone is connected to a different network, such as mobile data, or the computer's local IP address changes, the QR code will no longer connect to the application.

This is a limitation of the current development setup rather than the QR technology itself. A hosted version of the digital companion would be investigated during later development so that the QR codes can connect to the application regardless of the local network being used.

9. Next Development

The next stage of development could include:

- Testing the QR code with a physical printed or 3D-printed trap/card.
- Refining the mobile interface.
- Investigating a permanently hosted version of the digital companion.
- Creating additional physical components that can be identified through QR codes.
- Connecting additional game actions to physical components if required.
- Continuing to develop the physical board game and refining the trap mechanics through playtesting.

10. Technical PoC Result

The technical PoC was successful!!

The prototype demonstrated that a physical game component can connect to a Svelte-based digital companion using a QR code and trigger a digital game action. When the QR code was scanned using a mobile phone, the application identified the trap and automatically applied its game effect, reducing the Rat's health from 10 HP to 5 HP.

The test was successfully completed within the local development environment. Although the current implementation relies on a local network address, it demonstrates that the proposed physical to digital interaction is technically feasible. A hosted version of the digital companion can be investigated during later development so that QR coded physical components can function independently of the development environment.

